

AI & Machine Learning

142 questions · compiled from 2 files

1. What is the main goal of artificial intelligence?

1. To enable machines to perform tasks that typically require human intelligence
2. To replace all software with robots
3. To store data more cheaply
4. To eliminate the need for algorithms

Answer 1 · To enable machines to perform tasks that typically require human intelligence

2. Which of the following is an example of AI?

1. A calculator performing fixed arithmetic
2. A thermostat with no adaptation
3. A system that recognizes spoken commands
4. A printed spreadsheet

Answer 3 · A system that recognizes spoken commands

3. What best describes machine learning?

1. A method where systems learn patterns from data
2. A way to manually hard-code every decision
3. A database backup strategy
4. A type of computer monitor

Answer 1 · A method where systems learn patterns from data

4. Which statement is true about AI and machine learning?

1. Machine learning and AI are unrelated
2. AI is a subset of machine learning
3. Machine learning is one approach within AI
4. AI only refers to physical robots

Answer 3 · Machine learning is one approach within AI

5. In supervised learning, a model is trained using:

1. Only unlabeled data
2. Data with input-output pairs
3. Random internet text only
4. No examples at all

Answer 2 · Data with input-output pairs

6. Which task is a classification problem?

1. Predicting tomorrow's temperature exactly
2. Estimating house price
3. Assigning emails to spam or not spam
4. Compressing a file

Answer 3 · Assigning emails to spam or not spam

7. Which task is a regression problem?

1. Grouping customers by behavior
2. Predicting a continuous sales value
3. Sorting files alphabetically
4. Detecting whether an image contains a cat or dog only

Answer 2 · Predicting a continuous sales value

8. Unsupervised learning is most useful when:

1. You have labeled target outputs for every example
2. You want to discover structure in unlabeled data
3. You already know the exact answer rules
4. You need to compile source code

Answer 2 · You want to discover structure in unlabeled data

9. Which of the following is an unsupervised learning task?

1. Spam detection with labeled examples
2. Predicting exam scores from study hours
3. Clustering similar customers
4. Translating a sentence with a fixed dictionary only

Answer 3 · Clustering similar customers

10. What is reinforcement learning primarily based on?

1. Learning through rewards and penalties
2. Learning only from labeled tables
3. Sorting data by size
4. Removing duplicate rows

Answer 1 · Learning through rewards and penalties

11. In machine learning, a feature is:

1. The final prediction only
2. An input variable used by the model
3. A type of hardware accelerator
4. A chart title

Answer 2 · An input variable used by the model

12. In supervised learning, the label is:

1. The hidden layer size
2. The target output the model tries to predict
3. Any unused column
4. A random seed value

Answer 2 · The target output the model tries to predict

13. What is a model in machine learning?

1. A learned mapping from inputs to outputs or patterns
2. A spreadsheet formatting rule
3. A physical robot arm
4. A file compression method

Answer 1 · A learned mapping from inputs to outputs or patterns

14. What does training a model mean?

1. Deleting noisy rows permanently
2. Adjusting the model based on data
3. Drawing a dashboard for executives
4. Converting labels into charts

Answer 2 · Adjusting the model based on data

15. What is inference in AI?

1. The process of using a trained model to make predictions
2. The process of collecting raw data only
3. The process of installing Python
4. The process of renaming variables

Answer 1 · The process of using a trained model to make predictions

16. What is overfitting?

1. When a model performs well on training data but poorly on new data
2. When a model is too simple to learn any pattern
3. When a dataset has too many rows
4. When a chart uses too many colors

Answer 1 · When a model performs well on training data but poorly on new data

17. What is underfitting?

1. When a model memorizes noise in the training set
2. When a model is too simple to capture important patterns
3. When a model uses validation data
4. When a model has already been deployed

Answer 2 · When a model is too simple to capture important patterns

18. Which practice helps detect overfitting?

1. Evaluating only on the training set
2. Ignoring model errors
3. Testing on unseen validation or test data
4. Removing all features immediately

Answer 3 · Testing on unseen validation or test data

19. Why is a test set used?

1. To tune the model repeatedly during training
2. To estimate performance on unseen data
3. To increase the number of labels
4. To replace the training set entirely

Answer 2 · To estimate performance on unseen data

20. What is data leakage?

1. When information from outside the training process improperly influences the model
2. When a file is too large to upload
3. When labels are stored in CSV format
4. When a chart is missing a legend

Answer 1 · When information from outside the training process improperly influences the model

21. Which situation is an example of leakage?

1. Using future information that would not be available at prediction time
2. Normalizing training data
3. Splitting data into train and test sets
4. Removing duplicate records

Answer 1 · Using future information that would not be available at prediction time

22. What does generalization mean in machine learning?

1. The ability to perform well on new unseen data
2. The ability to memorize every training example
3. The ability to increase file size
4. The ability to avoid all preprocessing

Answer 1 · The ability to perform well on new unseen data

23. Which metric is commonly used for classification?

1. Accuracy
2. Mean squared error only
3. R-squared only
4. Variance inflation factor

Answer 1 · Accuracy

24. Which metric is commonly used for regression?

1. Precision
2. Recall
3. Mean absolute error
4. F1-score

Answer 3 · Mean absolute error

25. What does accuracy measure?

1. The average size of prediction errors
2. The proportion of correct predictions
3. The number of model parameters
4. The speed of data loading

Answer 2 · The proportion of correct predictions

26. When classes are imbalanced, accuracy alone can be misleading because:

1. It always becomes zero
2. A model can appear strong by predicting the majority class often
3. It cannot be computed
4. It only works for regression

Answer 2 · A model can appear strong by predicting the majority class often

27. Precision is most concerned with:

1. How many predicted positives were actually positive
2. How many actual positives were found
3. How close predictions are to a line
4. How many clusters were formed

Answer 1 · How many predicted positives were actually positive

28. Recall is most concerned with:

1. How many actual positives were correctly identified
2. How many predicted negatives were correct
3. How many features were dropped
4. How fast the model trains

Answer 1 · How many actual positives were correctly identified

29. What is the F1-score?

1. A measure of GPU memory
2. The harmonic mean of precision and recall
3. A clustering distance metric
4. A type of activation function only

Answer 2 · The harmonic mean of precision and recall

30. What is a confusion matrix used for?

1. To summarize classification prediction outcomes
2. To reduce image resolution
3. To store model weights
4. To normalize numeric columns

Answer 1 · To summarize classification prediction outcomes

31. Which algorithm is commonly associated with classification or regression?

1. Linear regression
2. K-means clustering
3. Apriori association rules only
4. Principal component analysis only

Answer 1 · Linear regression

32. Which algorithm is commonly used for clustering?

1. Decision tree
2. K-means
3. Linear regression
4. Logistic regression

Answer 2 · K-means

33. What is a decision tree?

1. A model that splits data based on feature conditions
2. A chart showing only time series
3. A database indexing method
4. A rule that always predicts the average

Answer 1 · A model that splits data based on feature conditions

34. What is the purpose of a validation set?

1. To tune model choices before final testing
2. To permanently replace the test set
3. To store production logs
4. To guarantee perfect accuracy

Answer 1 · To tune model choices before final testing

35. Why should training and test data be separated?

1. To make the dataset look larger
2. To estimate how the model performs on unseen examples
3. To avoid using labels
4. To remove all bias automatically

Answer 2 · To estimate how the model performs on unseen examples

36. What is bias in an AI system?

1. Any use of statistics
2. Systematic error or unfair skew in data or predictions
3. A required part of every chart title
4. The same thing as variance in every context

Answer 2 · Systematic error or unfair skew in data or predictions

37. Why is representative data important?

1. It helps the model learn patterns relevant to the real world
2. It guarantees zero errors
3. It removes the need for evaluation
4. It makes labels unnecessary

Answer 1 · It helps the model learn patterns relevant to the real world

38. What is a heuristic in AI?

1. A guaranteed optimal solution in all cases
2. A rule-of-thumb that helps guide problem solving
3. A hardware cooling method
4. A type of database join

Answer 2 · A rule-of-thumb that helps guide problem solving

39. In search problems, what does a heuristic ideally estimate?

1. The remaining cost to reach a goal
2. The number of rows in a dataset
3. The file extension of the input
4. The exact training labels

Answer 1 · The remaining cost to reach a goal

40. What is a constraint satisfaction problem?

1. A problem where variables must satisfy a set of rules
2. A problem that can only be solved with neural networks
3. A chart design problem
4. A method for compressing images

Answer 1 · A problem where variables must satisfy a set of rules

41. Sudoku is a classic example of:

1. Regression
2. Constraint satisfaction
3. Unsupervised clustering
4. Dimensionality reduction

Answer 2 · Constraint satisfaction

42. What is an intelligent agent?

1. An entity that perceives its environment and takes actions
2. A spreadsheet with formulas
3. A static PDF report
4. A file transfer protocol

Answer 1 · An entity that perceives its environment and takes actions

43. What does an agent's policy describe?

1. The mapping from situations to actions
2. The number of hidden layers
3. The color of the interface
4. The storage format of labels

Answer 1 · The mapping from situations to actions

44. What is deep learning?

1. A subset of machine learning using multi-layer neural networks
2. Any learning done on large files
3. A search algorithm for graphs only
4. A manual feature selection checklist

Answer 1 · A subset of machine learning using multi-layer neural networks

45. Why are neural networks powerful?

1. They can learn complex nonlinear relationships
2. They always require no data
3. They eliminate the need for evaluation
4. They only work for text

Answer 1 · They can learn complex nonlinear relationships

46. Which area focuses on enabling computers to understand human language?

1. Computer vision
2. Natural language processing
3. Operating systems
4. Network routing

Answer 2 · Natural language processing

47. Which area focuses on interpreting images and videos?

1. Computer vision
2. Reinforcement learning
3. SQL optimization
4. Spreadsheet auditing

Answer 1 · Computer vision

48. What is one major risk of deploying AI without human oversight?

1. The model may produce harmful or incorrect outputs without review
2. The model will stop using data types
3. The code will always fail to run
4. The dataset will automatically rebalance itself

Answer 1 · The model may produce harmful or incorrect outputs without review

49. Why is reproducibility important in machine learning?

1. It allows others to repeat and verify results
2. It guarantees the model is fair
3. It removes the need for documentation
4. It makes all models interpretable

Answer 1 · It allows others to repeat and verify results

50. What is the main purpose of feature engineering?

1. To improve learning by creating or selecting useful input variables
2. To increase monitor resolution
3. To replace evaluation metrics
4. To convert all tasks into clustering

Answer 1 · To improve learning by creating or selecting useful input variables

51. What is the difference between correlation and causation?

1. Correlation always proves one variable causes another
2. Causation means two variables are identical
3. Correlation indicates association, while causation means one factor directly influences another
4. There is no difference

Answer 3 · Correlation indicates association, while causation means one factor directly influences another

INTRODUCTION TO AI AND MACHINE LEARNING

52. Which description best matches artificial intelligence in an introductory sense?

1. Storing files in a database
2. Writing only mathematical proofs by hand
3. Building systems that perform tasks associated with human-like intelligence
4. Compressing images without analysis

Answer 3 · Building systems that perform tasks associated with human-like intelligence

INTRODUCTION TO AI AND MACHINE LEARNING

53. What is a key goal of many AI systems?

1. To eliminate all uncertainty from the world
2. To make decisions or predictions based on available information
3. To avoid using any data
4. To replace every human judgment automatically

Answer 2 · To make decisions or predictions based on available information

INTRODUCTION TO AI AND MACHINE LEARNING

54. In an introductory AI context what is a search problem?

1. A task where an agent explores possible states to reach a goal
2. A method for formatting spreadsheets
3. A way to compress text files
4. A type of network cable

Answer 1 · A task where an agent explores possible states to reach a goal

55. Why is state representation important in AI problem solving?

1. It removes the need for algorithms
2. It determines how the problem is modeled and what actions can be evaluated
3. It only affects screen layout
4. It guarantees the shortest solution

Answer 2 · It determines how the problem is modeled and what actions can be evaluated

56. Why are constraints useful in AI problems such as Sudoku?

1. They make every puzzle unsolvable
2. They reduce the space of possible solutions by enforcing rules
3. They replace all reasoning with guessing
4. They only improve graphics rendering

Answer 2 · They reduce the space of possible solutions by enforcing rules

57. What is the main benefit of constraint propagation?

1. It guarantees no backtracking is ever needed
2. It narrows possibilities early so later search becomes more efficient
3. It converts all problems into supervised learning
4. It stores all solutions permanently

Answer 2 · It narrows possibilities early so later search becomes more efficient

58. What does an agent do in AI?

1. It only stores data without acting
2. It perceives information and takes actions toward a goal
3. It always follows random behavior
4. It only works in robotics

Answer 2 · It perceives information and takes actions toward a goal

59. Which statement best describes machine learning?

1. A way to manually encode every rule in advance
2. A replacement for all software engineering
3. A set of methods that learn patterns from data to make predictions or decisions
4. A type of cloud storage

Answer 3 · A set of methods that learn patterns from data to make predictions or decisions

60. Why is machine learning often useful when explicit rules are difficult to write?

1. Because learning removes the need for data
2. Because explicit rules are never valid
3. Patterns can be inferred from examples instead of hand-coding every case
4. Because ML always explains itself perfectly

Answer 3 · Patterns can be inferred from examples instead of hand-coding every case

61. What is supervised learning primarily based on?

1. Learning only from unlabeled clusters
2. Learning by random exploration only
3. Learning without any data
4. Learning from labeled examples with known outputs

Answer 4 · Learning from labeled examples with known outputs

62. What is unsupervised learning mainly used for?

1. Training only with reward signals
2. Solving every optimization problem exactly
3. Replacing feature engineering completely
4. Finding structure or patterns in unlabeled data

Answer 4 · Finding structure or patterns in unlabeled data

63. What is reinforcement learning centered on?

1. Learning only from static labeled tables
2. Grouping similar records without feedback
3. Learning through actions and rewards over time
4. Memorizing all training examples exactly

Answer 3 · Learning through actions and rewards over time

64. How is reinforcement learning different from supervised learning?

1. It uses no objective at all
2. It cannot involve sequential decisions
3. It always requires fewer data
4. It learns from reward feedback on actions rather than direct labeled answers

Answer 4 · It learns from reward feedback on actions rather than direct labeled answers

65. What is training in machine learning?

1. Deleting incorrect rows manually
2. Adjusting model parameters based on data to improve performance
3. Converting code into machine language
4. Backing up the dataset to the cloud

Answer 2 · Adjusting model parameters based on data to improve performance

66. Why is a separate test set important?

1. It guarantees perfect fairness
2. It increases training accuracy automatically
3. It helps estimate how well the model generalizes to unseen data
4. It replaces the need for validation

Answer 3 · It helps estimate how well the model generalizes to unseen data

67. Which situation best indicates overfitting?

1. Low training and low test performance
2. High training performance but weak performance on unseen data
3. Equal performance on train and test
4. Stable results across multiple datasets

Answer 2 · High training performance but weak performance on unseen data

68. What is a feature in machine learning?

1. A final prediction label
2. An input variable used by the model to make predictions
3. A hardware accelerator
4. A chart title

Answer 2 · An input variable used by the model to make predictions

69. What is a label in supervised learning?

1. An unused metadata field
2. The target output the model is trying to predict
3. A type of hidden layer
4. A preprocessing script

Answer 2 · The target output the model is trying to predict

70. Why does data quality matter in ML?

1. Because models ignore the training data
2. Because data quality only affects storage cost
3. Because more data always fixes every issue
4. Poor or biased data can lead to poor or biased model behavior

Answer 4 · Poor or biased data can lead to poor or biased model behavior

71. What is classification?

1. Predicting only continuous numeric values
2. Predicting a category or class label
3. Sorting files by size
4. Reducing image resolution

Answer 2 · Predicting a category or class label

72. What is regression?

1. Predicting a continuous numeric value
2. Grouping unlabeled items into clusters
3. Choosing actions from rewards only
4. Removing outliers from a table

Answer 1 · Predicting a continuous numeric value

73. Which example is a classification task?

1. Estimating tomorrow's temperature in degrees
2. Predicting whether an email is spam or not spam
3. Compressing a video file
4. Sorting names alphabetically

Answer 2 · Predicting whether an email is spam or not spam

74. Which example is a regression task?

1. Assigning a news article to sports or politics
2. Detecting whether a login is fraudulent or not
3. Predicting house price from property features
4. Grouping customers by behavior without labels

Answer 3 · Predicting house price from property features

75. What is clustering?

1. Grouping similar data points without labeled outcomes
2. Predicting exact future sales values
3. Learning from reward signals only
4. Writing rules for every category manually

Answer 1 · Grouping similar data points without labeled outcomes

76. Why can feature selection improve a model?

1. It guarantees causality
2. Relevant inputs can reduce noise and improve generalization
3. It eliminates the need for evaluation
4. It always increases the number of classes

Answer 2 · Relevant inputs can reduce noise and improve generalization

77. What is deep learning generally associated with?

1. Only symbolic logic systems
2. Only spreadsheet formulas
3. Neural networks with multiple layers that learn complex patterns
4. Only database normalization

Answer 3 · Neural networks with multiple layers that learn complex patterns

78. What is natural language processing concerned with?

1. Designing only computer graphics
2. Managing server uptime
3. Helping computers work with human language text or speech
4. Encrypting network traffic

Answer 3 · Helping computers work with human language text or speech

79. Why is NLP challenging?

1. Because text contains no patterns
2. Because computers cannot store words
3. Human language is ambiguous context-dependent and variable
4. Because language tasks never use data

Answer 3 · Human language is ambiguous context-dependent and variable

80. What is computer vision about?

1. Generating only audio waveforms
2. Managing relational databases
3. Enabling systems to interpret images or visual data
4. Scheduling CPU tasks

Answer 3 · Enabling systems to interpret images or visual data

81. Why is evaluation important in AI and ML?

1. It guarantees deployment approval
2. It removes the need for training data
3. It checks whether a system actually meets its intended objective
4. It only matters after failure

Answer 3 · It checks whether a system actually meets its intended objective

82. What is a likely risk of deploying an unvalidated AI system?

1. Because deployment automatically improves accuracy
2. Because users will correct the model instantly
3. Because validation only affects documentation
4. It may behave unreliably or unfairly in real use

Answer 4 · It may behave unreliably or unfairly in real use

83. What does bias in AI often refer to?

1. A faster training process
2. Systematic error or unfair skew in data or model behavior
3. A type of GPU memory
4. A required step in optimization

Answer 2 · Systematic error or unfair skew in data or model behavior

84. Why should humans remain involved in many AI workflows?

1. AI systems are legally people
2. Humans only slow systems down
3. Human review is useful only for spelling
4. Human judgment is needed for oversight context and accountability

Answer 4 · Human judgment is needed for oversight context and accountability

85. What is the best reason to document assumptions in an AI project?

1. Documentation makes models train faster
2. Assumptions are only needed for marketing
3. Assumptions affect interpretation limitations and responsible use
4. Documentation replaces testing

Answer 3 · Assumptions affect interpretation limitations and responsible use

86. What is generalization in machine learning?

1. Memorizing every row in the dataset
2. Running the same code on different laptops
3. Performing well on new unseen data rather than only training examples
4. Combining all models into one file

Answer 3 · Performing well on new unseen data rather than only training examples

87. What is a common purpose of a validation set?

1. To replace the training set entirely
2. To tune choices before final testing
3. To store production logs
4. To label raw data automatically

Answer 2 · To tune choices before final testing

88. Why is data leakage harmful?

1. It always reduces training speed only
2. It improves fairness automatically
3. It makes evaluation overly optimistic by exposing information that would not be available in real use
4. It is the same as regularization

Answer 3 · It makes evaluation overly optimistic by exposing information that would not be available in real use

89. What is an example of an AI system using search rather than learning?

1. A spam filter trained on labeled emails
2. A recommendation model trained on clicks
3. A Sudoku solver exploring valid assignments under constraints
4. A clustering algorithm grouping customers

Answer 3 · A Sudoku solver exploring valid assignments under constraints

90. What is one reason introductory AI courses include puzzles like Sudoku?

1. They are the main industrial use of AI
2. They illustrate reasoning search and constraints clearly
3. They require no algorithms
4. They replace all machine learning concepts

Answer 2 · They illustrate reasoning search and constraints clearly

91. Why should students distinguish AI from machine learning?

1. They are unrelated fields
2. AI only means robotics while ML means databases
3. Machine learning is one approach within the broader field of AI
4. Machine learning always excludes data

Answer 3 · Machine learning is one approach within the broader field of AI

92. What is the role of reward in reinforcement learning?

1. It labels every example in advance
2. It signals how desirable an action outcome is
3. It groups similar records
4. It stores model parameters

Answer 2 · It signals how desirable an action outcome is

93. What is a practical first step before choosing an ML model?

1. Pick the most complex model immediately
2. Deploy before evaluation
3. Define the problem objective and available data clearly
4. Ignore the target variable

Answer 3 · Define the problem objective and available data clearly

94. Why might a simpler model sometimes be preferred over a more complex one?

1. Because complex models never work
2. Because simple models require no data
3. Because simplicity guarantees perfect accuracy
4. It can be easier to interpret maintain and generalize adequately

Answer 4 · It can be easier to interpret maintain and generalize adequately

95. What is the main purpose of preprocessing data?

1. To guarantee causal conclusions
2. To eliminate all missing values automatically
3. To prepare inputs so the model can learn from them more effectively
4. To replace evaluation metrics

Answer 3 · To prepare inputs so the model can learn from them more effectively

96. Why is reproducibility valuable in AI and ML work?

1. It makes models larger
2. It removes the need for documentation
3. It helps others verify repeat and trust the results
4. It guarantees commercial success

Answer 3 · It helps others verify repeat and trust the results

97. What is generative AI primarily designed to do?

1. Only classify existing records
2. Create new content such as text images or code based on learned patterns
3. Only sort structured tables
4. Only compress files

Answer 2 · Create new content such as text images or code based on learned patterns

98. Which course area most directly supports generative AI foundations in your program?

1. Cloud Fundamentals Fluency
2. Introduction to SQL only
3. Generative AI Fluency
4. AWS Networking only

Answer 3 · Generative AI Fluency

99. What is a large language model best described as?

1. A spreadsheet with macros
2. A database query engine
3. A graphics card driver
4. A model trained on large amounts of text to generate or transform language

Answer 4 · A model trained on large amounts of text to generate or transform language

100. Why can LLMs produce convincing but incorrect answers?

1. Because they always retrieve from verified databases
2. Because they only answer with copied text
3. Because they refuse ambiguous prompts
4. They generate likely sequences of text rather than guaranteeing truth

Answer 4 · They generate likely sequences of text rather than guaranteeing truth

101. What is prompting?

1. Compressing the model before use
2. Providing instructions or context to guide model output
3. Labeling a dataset for training
4. Deploying a model to production

Answer 2 · Providing instructions or context to guide model output

102. Why does prompt specificity matter?

1. Specific prompts make the model smaller
2. Clear instructions improve relevance structure and usefulness of the output
3. Specificity removes all hallucinations
4. Specific prompts are only useful for images

Answer 2 · Clear instructions improve relevance structure and usefulness of the output

103. What is a strong reason to include examples in a prompt?

1. Because examples force factual correctness
2. Because examples disable creativity
3. Examples demonstrate the desired pattern format or style
4. Because examples are only for beginners

Answer 3 · Examples demonstrate the desired pattern format or style

104. What is zero-shot prompting?

1. Training a model from scratch
2. Asking the model to perform a task without giving worked examples
3. Using zero data in production
4. Removing all instructions from a prompt

Answer 2 · Asking the model to perform a task without giving worked examples

105. What is few-shot prompting?

1. Using only a few model parameters
2. Training on a tiny dataset only
3. Providing a small number of examples to guide the model
4. Running the model for a few seconds

Answer 3 · Providing a small number of examples to guide the model

106. Why can few-shot prompting improve output quality?

1. Because it changes the model architecture
2. Because it guarantees no bias
3. It gives the model a clearer pattern to imitate for the task
4. Because it replaces evaluation

Answer 3 · It gives the model a clearer pattern to imitate for the task

107. What is retrieval-augmented generation or RAG?

1. A way to compress prompts
2. A type of reinforcement learning
3. A method that combines retrieval of relevant information with generation
4. A hardware scaling technique

Answer 3 · A method that combines retrieval of relevant information with generation

108. Why is RAG useful?

1. It removes the need for any documents
2. It can ground responses in retrieved context rather than relying only on model memory
3. It guarantees perfect citations
4. It only works for images

Answer 2 · It can ground responses in retrieved context rather than relying only on model memory

109. What problem does RAG most directly help reduce?

1. Basic arithmetic speed
2. Screen resolution issues
3. Network encryption errors
4. Unsupported answers caused by missing or outdated context

Answer 4 · Unsupported answers caused by missing or outdated context

110. What are embeddings commonly used for in modern GenAI systems?

1. Encrypting files at rest
2. Representing meaning in vector form for similarity search and retrieval
3. Rendering charts in dashboards
4. Scheduling GPU jobs

Answer 2 · Representing meaning in vector form for similarity search and retrieval

111. What is vector search used for?

1. Sorting rows alphabetically
2. Training a model from scratch
3. Finding semantically similar items based on embeddings
4. Reducing image brightness

Answer 3 · Finding semantically similar items based on embeddings

112. Why are tokens important in LLM systems?

1. Because tokens are only used in databases
2. Because tokens guarantee factuality
3. They are the units text is broken into for processing and cost accounting
4. Because tokens are the same as embeddings

Answer 3 · They are the units text is broken into for processing and cost accounting

113. What is fine-tuning in a generative AI context?

1. Deleting model weights to save storage
2. Adapting a pretrained model further for a task or behavior
3. Only changing the prompt wording
4. Converting text into SQL tables

Answer 2 · Adapting a pretrained model further for a task or behavior

114. What is a practical reason to fine-tune a model?

1. To avoid all evaluation
2. To better align it with a domain style task or behavior
3. To eliminate the need for data governance
4. To make retrieval unnecessary in every case

Answer 2 · To better align it with a domain style task or behavior

115. Why might prompt engineering be preferred before fine-tuning?

1. Because fine-tuning never works
2. Because prompting changes the training data
3. Because prompt engineering guarantees domain expertise
4. It is often faster cheaper and easier to iterate initially

Answer 4 · It is often faster cheaper and easier to iterate initially

116. What is an agent in the generative AI ecosystem?

1. A static PDF report
2. A database backup process
3. A system that uses a model to reason plan or act with tools toward goals
4. A type of image filter

Answer 3 · A system that uses a model to reason plan or act with tools toward goals

117. Why are tools important for AI agents?

1. Because tools replace prompts
2. Because tools make models deterministic
3. Because tools are only for debugging
4. They let agents interact with external systems or data sources

Answer 4 · They let agents interact with external systems or data sources

118. What is a key concern with agentic systems?

1. Because agents cannot use language
2. Because agents never need evaluation
3. Because agents always outperform simple workflows
4. They can take incorrect actions if goals context or safeguards are weak

Answer 4 · They can take incorrect actions if goals context or safeguards are weak

119. What is hallucination in generative AI?

1. A model refusing to answer
2. A hardware overheating issue
3. Confidently producing content that is false unsupported or fabricated
4. A type of vector database

Answer 3 · Confidently producing content that is false unsupported or fabricated

120. What is a good mitigation for hallucinations?

1. Ask shorter questions only
2. Never use examples
3. Ground the model with reliable context and verify important claims
4. Increase font size in the interface

Answer 3 · Ground the model with reliable context and verify important claims

121. Why is human review still important in generative AI workflows?

1. Because humans only need to review spelling
2. Because AI outputs are legally binding
3. Because review is unnecessary after deployment
4. Generated outputs may be plausible yet inaccurate incomplete or risky

Answer 4 · Generated outputs may be plausible yet inaccurate incomplete or risky

122. What is model adaptation generally about?

1. Changing monitor settings
2. Moving files between folders
3. Adjusting a foundation model for a new task or domain
4. Replacing prompts with spreadsheets

Answer 3 · Adjusting a foundation model for a new task or domain

123. What is one benefit of structured output from a model?

1. It guarantees truthfulness
2. It makes downstream processing and validation easier
3. It removes the need for prompts
4. It only helps image generation

Answer 2 · It makes downstream processing and validation easier

124. Why is evaluation important for generative AI systems?

1. Because generation quality is obvious from one example
2. Because evaluation only matters for training loss
3. Quality must be assessed for accuracy usefulness safety and consistency
4. Because users never notice failures

Answer 3 · Quality must be assessed for accuracy usefulness safety and consistency

125. What is context window referring to in LLM usage?

1. The size of the browser window
2. The amount of text the model can consider at one time
3. The number of users on a team
4. The speed of internet access

Answer 2 · The amount of text the model can consider at one time

126. Why can long context still require careful prompt design?

1. Because long context disables generation
2. Important details can be diluted if instructions and evidence are not organized well
3. Because context length guarantees reasoning
4. Because models ignore recent text

Answer 2 · Important details can be diluted if instructions and evidence are not organized well

127. What is the main value of systematizing prompts in production workflows?

1. Because it makes models open source
2. Because it removes all governance needs
3. It improves consistency repeatability and maintainability across use cases
4. Because it guarantees zero latency

Answer 3 · It improves consistency repeatability and maintainability across use cases

128. What is multimodal generative AI?

1. AI that only processes SQL queries
2. AI that only runs on mobile devices
3. AI that can work with more than one type of input or output such as text and images
4. AI that cannot use language

Answer 3 · AI that can work with more than one type of input or output such as text and images

129. What is a common enterprise use of generative AI?

1. Replacing all security controls
2. Drafting summarizing searching or assisting with knowledge work
3. Eliminating the need for data ownership
4. Guaranteeing legal compliance

Answer 2 · Drafting summarizing searching or assisting with knowledge work

130. Why should sensitive data be handled carefully in GenAI workflows?

1. Because sensitive data improves hallucinations
2. Because models cannot read sensitive text
3. Privacy security and governance obligations still apply to prompts and outputs
4. Because governance only matters after launch

Answer 3 · Privacy security and governance obligations still apply to prompts and outputs

131. What is prompt refinement?

1. Retraining the model from scratch
2. Improving a prompt after reviewing output to better match the goal
3. Reducing the dataset size
4. Deleting context to save tokens

Answer 2 · Improving a prompt after reviewing output to better match the goal

132. What is a useful way to ask for better model output?

1. Use vague requests repeatedly
2. Specify audience format constraints and success criteria
3. Avoid giving any context
4. Never mention the task objective

Answer 2 · Specify audience format constraints and success criteria

133. Why is decomposition helpful in generative AI tasks?

1. Because models cannot answer short tasks
2. Because decomposition removes the need for validation
3. Because one-step prompts are always wrong
4. Breaking a complex task into steps often improves control and reliability

Answer 4 · Breaking a complex task into steps often improves control and reliability

134. In a generic enterprise-AI sense what could an AI Refinery concept imply?

1. A physical machine for cooling servers
2. A managed process for transforming raw AI components into usable workflows or outputs
3. A spreadsheet macro for formatting cells
4. A firewall rule for blocking prompts

Answer 2 · A managed process for transforming raw AI components into usable workflows or outputs

135. If AI Refinery is treated generically rather than as a named course feature what is the safest interpretation?

1. A guaranteed Udacity product capability
2. A conceptual pipeline for preparing orchestrating or improving AI outputs
3. A specific database engine
4. A replacement for model evaluation

Answer 2 · A conceptual pipeline for preparing orchestrating or improving AI outputs

136. Why is it important not to overclaim platform-specific features like AI Refinery when they are not explicitly taught?

1. Because generic concepts are never useful
2. Because enterprise tools cannot relate to AI
3. Accurate scope prevents misleading learners about what the curriculum directly covers
4. Because named features always behave identically

Answer 3 · Accurate scope prevents misleading learners about what the curriculum directly covers

137. What could a generic Stream specific Assets view most reasonably refer to?

1. A video player setting
2. A GPU overclocking panel
3. A filtered view of resources relevant to a particular workflow stream or use case
4. A method for deleting embeddings

Answer 3 · A filtered view of resources relevant to a particular workflow stream or use case

138. Why might a stream-specific assets view be useful in an AI workflow?

1. It guarantees model accuracy
2. It helps users focus on the most relevant documents models or components for a task
3. It replaces access control
4. It prevents all duplication

Answer 2 · It helps users focus on the most relevant documents models or components for a task

139. How should learners treat concepts like stream-specific asset organization if not explicitly listed in the outline?

1. As mandatory exam features
2. As transferable workflow ideas rather than confirmed course-specific product features
3. As proof of hidden modules
4. As replacements for retrieval systems

Answer 2 · As transferable workflow ideas rather than confirmed course-specific product features

140. What is the relationship between generative AI and later agent courses in your program?

1. They are unrelated topics
2. Agents replace all language models
3. GenAI is only a prerequisite for SQL
4. Later courses build on GenAI foundations by adding tools workflows memory and orchestration

Answer 4 · Later courses build on GenAI foundations by adding tools workflows memory and orchestration

141. Why are guardrails important in generative or agentic systems?

1. They increase randomness
2. They remove the need for prompts
3. They help reduce unsafe off-target or policy-violating behavior
4. They only improve visual design

Answer 3 · They help reduce unsafe off-target or policy-violating behavior

142. What is the best mindset for using generative AI in professional settings?

1. Treat it as an infallible autonomous expert
2. Use it only when no goals are defined
3. Assume every answer is production-ready
4. Treat it as a powerful assistant whose outputs require context verification and judgment

Answer 4 · Treat it as a powerful assistant whose outputs require context verification and judgment